ETHAN TREGIDGA

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Personal Website

SUMMARY

Machine learning researcher with expertise in deep learning, simulation-based inference, and astrophysics.
Strong communicator with experience presenting to industry, academia, and public audiences.

EDUCATION

2023-2027 Lausanne, CH
EPFL
PhD in Astrophysics with Machine Learning (in-progress)
Supervisors: Prof. Jean-Paul Kneib, Dr. David Harvey

2022-2023 Cambridge, US
Harvard University
Visiting Master in Astrophysics
Best Project By Final Year Research
Supervisors: Dr. James Steiner, Dr. Cecilia Garraffo

2019-2023 Southampton, UK
Southampton University
MPhys Astrophysics
First-Class Honours, 88% final grade (Most Outstanding Performance)

SKILLS

Developer: Python (PyTorch, NumPy, SciPy, Matplotlib, Multiprocessing, Django), JavaScript, HTML, PostgreSQL, Git, LaTeX, Linux

Advanced: C++, Rust, Bash

LANGUAGE

English (Native)
French (B1-B2)
Spanish (B1-B2)
German (A2)

PUBLICATIONS

F. Agurto-Sepúlveda, **et al.**, 2026, preprint
E. Tregidga, et al., 2026, A&A
F. Redmen, **E. Tregidga**, et al., 2025, ML4PS NeurIPS
D. Harvey, **et al.**, A&A, 2025
F. Vecchi, **et al.**, A&A, 2025
M. Aubin, **et al.**, Springer, 2025
E. Tregidga, et al., 2024, MNRAS

EXPERIENCE

October 2023 – Present Lausanne, CH
École Polytechnique Fédérale de Lausanne (EPFL)
PhD Research: Investigating Self-Interacting Dark Matter in Galaxy Clusters using Interpretable Machine Learning
- Designed a neural network that aligns dark matter simulations with observations in an interpretable latent space, identifying observation-simulation misalignment, improving trust in scientific machine learning.
- Built a PyTorch package to simplify and accelerate the building and prototyping of neural networks.
- Led several master and bachelor student projects to use foundation and denoising models in astronomy, along with teaching problem classes.

September 2022 – June 2023 Cambridge, US
Center for Astrophysics | Harvard & Smithsonian
Masters Research: Accelerating Black Hole Parameter Inference with Physicalised Autoencoders
- Developed a physicalised-latent-space autoencoder for black hole property prediction, achieving comparable performance while reducing computational time by orders of magnitude.

June – August 2021 Reading, UK
Thales UK Research & Technology
Improving Satellite Object Detection with Capsule Neural Networks
- Designed YOLO-CapsNet, combining capsule networks with YOLO, for improved real-time object detection in satellite and aerial imagery.

PROJECTS

September 2022 – Present
Center for Astrophysics | Harvard & Smithsonian
Interactive Data Archive Website for NICER X-Ray Telescope
- Lead developer for a data visualiser and analyser website for the neutron star interior composition explorer (NICER) X-ray telescope.
- Migrated website to PostgreSQL for improved website responsiveness.

June – August 2024
Center for Astrophysics | Harvard & Smithsonian
Combining X-Ray Spectroscopic Imaging Telescopes with Neural Networks to Achieve Super-Resolution
- Prototyped a super-resolution point cloud autoencoder to encode two X-ray telescopes' data cubes, generating the combined information.

November 2021 – May 2022
LaunchUK Nanosat Design Competition
Reducing UK Greenhouse Emissions with Nanosat Peatland Mapping
- Top-5 nationwide for designing a nano-satellite to use GPS signals-of-opportunity to detect greenhouse emissions from peatlands.

October 2019 – September 2022
UKSEDS Olympus Rover Trials
Martian Rover with Automatic Pathing and Sample Fetch
- First nationwide in 2020 & 2021, including best mission design and presentation. Head of software, developed locomotion and web control.